

Organisms - G8 Report

Vedha Avali, Mihnea-Stefan David, Evelyn Yu

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1 Introduction

In this project, we study the problem of designing adaptive artificial agents in the Organisms simulation environment, where the objective is to maximize long-term survival and population stability on a dynamic grid-based world. Each organism occupies a cell on an $m \times n$ toroidal grid and operates under strict energy constraints, gaining energy by consuming food and losing energy through movement, inactivity, and reproduction. The environment is also defined by a number of parameters:

- p : probability of spontaneous appearance of food
- q : probability of food doubling
- s : energy consumed in staying put
- v : energy consumed in moving or reproducing
- u : energy per unit of food
- M : maximum energy per organism
- K : maximum food units per cell

Some of the variables: s, v, u, M , and K are known to all organisms. The rest, as well as m and n , are not. Organisms must continuously balance exploration, resource consumption, and replication while competing with other organisms for limited food resources. The problem combines elements of decentralized decision-making, resource optimization, population control, and competitive multi-agent interaction under partial observability.

Our solution focuses on building a single adaptive organism capable of performing robustly across a wide range of environmental conditions, rather than relying on multiple specialized strategies. We designed a general decision framework in which the organism follows the same core behavioral rules: movement, reproduction, and waiting, but dynamically adjusts its decision thresholds based on environmental estimates and lifecycle phase.

To support adaptability, organisms estimate the food spawn probability from local observations, categorize environments into discrete probability buckets, and share this information with nearby allies through encoded external states. We further incorporate confidence-weighted voting mechanisms, inheritance-based knowledge bootstrapping for offspring, and population-control rules that regulate replication based on age, resource availability, and competitive pressure. These mechanisms enable coordinated behavior, efficient resource utilization, and resilience in both isolated and competitive settings. The remainder of this report details the design principles, adaptive strategies, and experimental results that demonstrate the effectiveness of our approach.

2 Initial Insights and Observations

2.1 Limited Information Landscape

After being presented with the problem statement, one of the most notable characteristics of the simulation environment is the extremely limited information available to each player at every round. At every round, an organism only knows

- Whether there is food or not on a neighboring square (but not how much food).

- Whether there is another organism on a neighboring square. If there is, then the external state of the neighboring organism is also available.
- The amount of remaining food on the organism's current cell.
- The amount of energy currently possessed by the organism.

Organisms cannot "see" any of the other locations on the board other than the four squares directly adjacent to their own. Additionally, a lot of important fixed environment parameters are not available to the organism.

Since the information given to each organism at every round is quite limited, it is hard to make well-informed, fine-grained decisions just based on the immediate environment. This leads one to conclude that even a fairly basic player may be quite successful, as there is an upper bound to the amount of computation about the inputs that is possible.

2.2 Single-Player Observations

In the single-player environment, let us first consider the matter of extinction. We initially tested a very basic player that roughly does the following:

- If there is food on a nearby cell, it moves on to the nearby cell.
- If its energy is above a certain percentage of its maximum energy, it reproduces.
- Some percentage of the time, even when there is no food nearby, it moves in a random direction.

We began with some arbitrary parameters for each of the random percentages. Even with this extremely basic model, in single-player environments, the organism did not die out except when p and q were set to be quite low (0.001 and 0.002, respectively). The organism usually experienced population cycles similar to those seen in the real world: in the beginning, it would populate and reproduce very aggressively, but eventually over-population would thin the population down again because of lack of food, and the cycle would continue. But the starvation period never resulted in extinction unless the environment was especially unfriendly.

We executed an iterative process of changing the values of the random percentages used to decide moving and replication and eventually created a few models that, even in very harsh food environments, generally did not go extinct (80-90% survival).

We will now move on to population density. In the single player environment, pure survival is not necessarily a very good metric of success. As described, even a fairly rudimentary player was able to survive in many environments if there were no competitors. The next logical metric is to try and maximize the surviving population count. Let us examine what the goal for this should be.

It is impossible to constantly support a population size of $m \times n$. A square with an organism on it will not randomly generate food and the food on it will not double. However, imagine that there are x free squares on the board. Then the population is $m \times n - x$. Let the average food per square be y . Let amount of food being generated in every timestep be V . Then

$$E[V] = mnx(yq + p)$$

Let's say that the population is stable when the amount of food being generated in any timestep is the same as the minimum of energy spent by all of the organisms in that timestep (generally 1/organism). Then we need

$$\begin{aligned} mn - x &\geq mnxu(yq + p) \\ mn &\geq mnxu(yq + p) \\ 1 &\geq xu(yq + p) \\ \frac{1}{u(yq + p)} &\geq x \end{aligned}$$

In the best case scenario, $y = K$. In our default scenario, this gives us

$$x \leq \frac{1}{100(80 * 0.02 + 0.01)} < 1$$

So in the best case scenario, the number of free squares necessary is less than 1. Of course, this is not very realistic: organisms have to move, and not every square will always have the maximum food on it. However, it does tell us

that in friendly environments, we should be aiming for more than pure survival; we could technically populate a very significant portion of the board.

But how do we get to this complete food population? In the beginning of an organism’s life, if it is above a certain percentage of its maximum energy, it waits before moving/reproducing/eating any food. This allows the food to grow abundantly (and exponentially), providing the organism with a lush field to grow in. After experimenting with this strategy, we found that our organisms had significantly better final population counts when they waited for some percentage of their early life.

2.3 Multi-Player Complications

In competitive play, an organism’s strategy has to change. We begin by assessing the challenges. Firstly, the organism suffers from the prisoner’s dilemma. Whilst it may be possible for many populations to exist in a given environment, if your competitor’s strategy is to wipe you out, by being more cordial, you will surely die out. Secondly, as previously stated, waiting for food to grow is one key to large population counts. However, this is impossible in a competitive environment: your competitors will eat the food that you are trying so hard to grow.

In a multi-player environment, it seems that early aggression is necessary. The ideal strategy seems to be spreading as far as you can across the environment as fast as possible, you deprive other competitors of food, and once they die out, dealing with the now-single-player environment as you ordinarily would.

A particularly acute difficulty that occurs in food-dense environments is that the total density of organisms is quite high, so it is not uncommon that some of your players may be surrounded on all four sides by other organisms. In this situation, they will eventually starve and die. There is no available growing or spawning food and they cannot escape in search of food. Avoiding this type of situation not only requires aggressive reproduction, but coordination among a species to spread out with enough space that food can grow, but tightly enough that other organisms cannot intrude. This is an especially difficult question in the presence of organisms that imitate the external state of others.

3 Strategies and Concepts

This section presents the high-level design principles that guided the development of our organism. Rather than focusing on implementation details, we summarize the main strategic ideas, behavioral models, and conceptual foundations underlying our approach. The topics introduced here are examined in greater theoretical and technical depth in the following section, where we provide detailed explanations of the mechanisms involved.

Our primary goal was to design an adaptive organism capable of surviving and remaining competitive across a wide variety of environments.

3.1 Adaptive Design Philosophy and Behavioral Model

During the early stages of development, we considered designing multiple specialized organisms optimized for specific environmental conditions. Although effective in controlled settings, these organisms performed poorly when parameters changed, making them unreliable in diverse and unpredictable scenarios.

To address this limitation, we adopted a design philosophy based on a *single adaptive brain*. Instead of switching between specialized strategies, our organism maintains a unified behavioral framework and adjusts its behavior dynamically according to environmental conditions. This approach improves robustness and reduces the risk of failure in unfamiliar environments.

Our design prioritizes **adaptability over dominance**. Rather than maximizing short-term population growth, we aim to ensure stable long-term survival across varying conditions. Overly aggressive replication can quickly exhaust resources and lead to population collapse, making sustainable behavior more effective.

A key principle of our approach is that the organism follows the same core decision-making model in all environments. Adaptation is achieved by adjusting internal **behavioral thresholds** rather than changing the underlying logic. These thresholds determine when an organism chooses to move, reproduce, or conserve energy, enabling smooth transitions between conservative and aggressive behaviors without requiring specialized rule sets.

3.2 Environmental Understanding and Uncertainty Management

Effective adaptation requires organisms to infer important properties of their environment, particularly food availability, which directly influences energy gain, reproduction, and survival. A key parameter is the food spawn probability p , which organisms cannot observe directly and must instead approximate through local observations gathered over time.

This estimation process is challenging due to partial observability. Organisms can sense only nearby cells and lack a global view of the grid, meaning environmental knowledge must emerge from limited and noisy local data.

Because precise numerical estimates are highly sensitive to noise, relying on exact values may lead to unstable behavior. To improve robustness, we categorize environments into a small number of discrete probability buckets (e.g., Extra Low, Low, Moderate, High). By reacting to broader environment categories rather than exact measurements, organisms maintain more consistent behavior despite estimation errors.

In addition, each estimate is associated with a confidence level reflecting the reliability of the collected observations. Estimates based on limited or inconsistent data receive lower confidence, encouraging cautious behavioral adjustments. This prevents organisms from overreacting to weak signals and promotes more stable adaptation under uncertainty.

Figure 1: *Environment Probability Buckets Used for Behavioral Adaptation*

Range	Bucket
$p < 0.005$	EXTRA_LOW
$p < 0.01$	VERY_LOW
$p < 0.05$	LOW
$p < 0.10$	MODERATE
otherwise	HIGH

3.3 Social Communication and Collective Decision-Making

Since individual organisms operate with limited local information, relying solely on personal observations may lead to inaccurate environmental assessments. To improve estimation quality, organisms share information with nearby peers using the external state mechanism provided by the simulator.

The external state acts as a compact communication channel through which organisms broadcast key information about their internal beliefs. This includes an encoded species identifier (TAG) that allows organisms to recognize members of the same species, as well as summaries of environmental estimates.

Species identification is important for ensuring reliable cooperation. Information received from organisms with matching identifiers is treated as trustworthy, while signals from unknown organisms may be ignored or down-weighted to reduce the risk of misleading or adversarial information.

Rather than relying exclusively on local estimates, organisms combine information from neighbors through a weighted voting process. Each received estimate contributes proportionally to its reliability, allowing the organism to aggregate multiple perspectives into a more accurate global assessment. Shared information is stored locally over time, allowing organisms to aggregate observations across multiple encounters rather than relying on momentary interactions.

This distributed consensus mechanism improves robustness by reducing sensitivity to local noise and individual estimation errors. As a result, organisms benefit from collective intelligence, enabling more stable and informed decision-making than would be possible through isolated observations alone.

3.4 Lifecycle-Aware Population Control

Organism behavior evolves over time to balance population growth with long-term survival. Optimal strategies vary across the lifecycle, with early stages favoring expansion and later stages emphasizing stability and resource conservation.

During early life, organisms behave more aggressively to establish a presence in the environment. Movement and reproduction are encouraged to explore the grid, secure food sources, and expand population reach, improving

access to resources and overall resilience.

As organisms mature, their strategy gradually shifts toward conservation. Instead of rapid expansion, older organisms prioritize energy preservation, risk avoidance, and population stability. This shift reduces resource depletion and lowers the risk of collapse caused by over-aggressive replication.

This transition is enabled by adjusting behavioral thresholds according to lifecycle stage. Different age phases use distinct parameter settings, allowing the same behavioral framework to support both expansion-oriented and conservation-oriented strategies.

Reproduction is further regulated through explicit population control mechanisms. Offspring limits prevent excessive replication and overcrowding, ensuring sustainable growth. By moderating reproduction frequency, organisms avoid resource exhaustion and maintain long-term ecological balance.

3.5 Knowledge Inheritance

To improve early survival and reduce the cost of learning from scratch, offspring organisms inherit selected information from their parent. Instead of starting with no prior knowledge, new organisms receive a “warm start” that allows them to make more informed decisions immediately after birth.

This inheritance mechanism functions as a simplified form of DNA bootstrapping, where essential environmental knowledge and behavioral parameters are transferred across generations. By preserving relevant experience, offspring can avoid inefficient exploratory phases and adapt more quickly to their surroundings.

Knowledge transfer improves population resilience by enabling faster stabilization and reducing vulnerability during early life stages. As a result, the population benefits from cumulative experience across generations rather than relying solely on individual learning.

3.6 Competitive Adaptations

While many behavioral mechanisms are designed for general survival, competitive environments introduce additional challenges. The presence of other organisms creates indirect interference through resource depletion, movement blocking, and environmental distortion, making decision-making more complex.

In such settings, environmental estimation becomes more difficult due to biased observations. Local measurements may reflect not only natural environmental properties but also the impact of competing organisms, leading to inaccurate assessments of resource availability.

To address these challenges, we introduce adjustments that account for competitive pressure. Behavioral thresholds are fine-tuned to reduce over-aggressive actions and to promote more cautious resource usage when interference is likely. These adaptations help organisms remain stable and avoid risky decisions driven by misleading environmental signals.

By incorporating competition-aware adjustments, the organism maintains more reliable behavior even in crowded and adversarial settings.

4 Implementation

4.1 System Overview

The organism is implemented as a modular decision-making system that processes local observations, updates internal state variables, and selects an action at each simulation step. The overall architecture follows a perception–processing decision pipeline designed to integrate environmental estimation, social information, and lifecycle-dependent behavioral control.

At every turn, the organism receives information about nearby food, neighboring organisms, and its current energy level. These inputs are first converted into structured internal representations of neighboring cells, allowing the organism to reason about its local environment.

The system then performs a sequence of processing stages. It updates environmental statistics used for estimating food spawn probability, integrates information received from nearby organisms, and computes a consensus-based

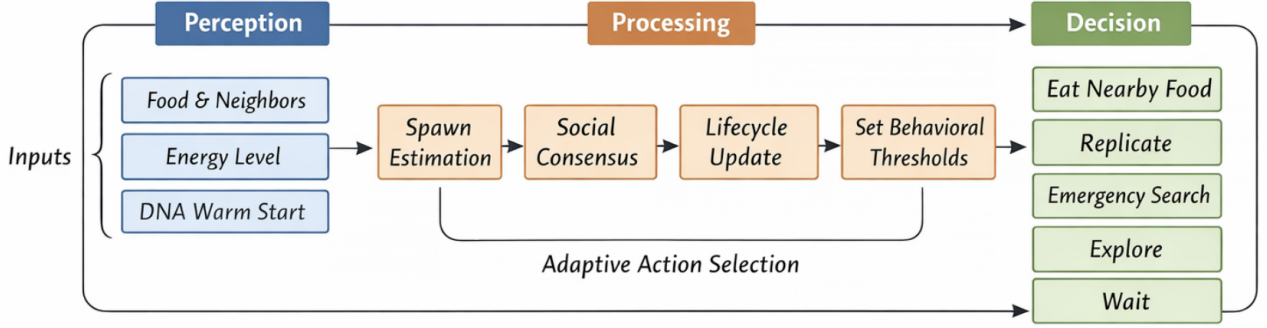


Figure 2: Overview of the organism decision-making pipeline. Local observations are processed through environmental estimation, social consensus, and lifecycle-dependent threshold adjustments before the decision engine selects an action.

assessment of environmental conditions. This assessment determines the current environment category, which directly influences internal behavioral thresholds controlling movement, reproduction, and energy conservation.

Lifecycle variables, such as age and offspring count, are updated continuously and further modulate behavioral parameters to ensure balanced population growth and long-term stability. In parallel, inherited information encoded in the organism’s DNA is decoded during early life stages to provide a warm start for environmental estimation.

Finally, the decision engine selects an action using a prioritized set of behavioral rules. Immediate survival actions, such as consuming nearby food, take precedence, followed by reproduction opportunities, emergency search behaviors under low energy, exploration strategies, and passive waiting when no beneficial action is available.

This modular organization allows each subsystem to contribute independently to the final behavior while maintaining a coherent global strategy. As a result, the organism can adapt its decisions dynamically based on environmental conditions, social context, and lifecycle stage.

4.2 Initial Parameter Tuning

To begin, we needed to develop a baseline understanding of what “ideal” qualities of an organism might be in specific environments. We designed three different tunable thresholds that would be modifiable in our basic organism.

Threshold Semantics. The three thresholds have distinct roles:

- **waitThreshold** controls how long the organism remains in a waiting-oriented phase. Larger values encourage more conservative behavior by delaying active exploration. Expressed as a percentage of maximum energy, w . When an organism is first born, it will wait without moving until it has gotten to $(100 - w)$ percent of its total energy.
- **replThreshold** specifies the minimum energy level required for reproduction, expressed as a percentage of the maximum energy M . Larger values make reproduction more selective.
- **moveThreshold** determines the probability of entering exploration mode. Larger values encourage movement and expansion. Expressed as a percentage. Each round at which the organism has no adjacent food, there is **moveThreshold** probability that the organism will move randomly onto a free neighboring square.

We also toggled a boolean parameter called **modesty**. If the parameter was true, then an organism would only reproduce onto cells with food, otherwise they would not reproduce, even if they met their reproduction threshold.

Together, these parameters define the organism’s aggressiveness profile. High movement probability combined with a low reproduction threshold corresponds to expansion-oriented behavior, while high waiting and reproduction thresholds lead to more cautious strategies.

We began by creating an organism that used a genetic algorithm to slowly adapt over the course of individual games. Each reproduction, the thresholds had a small probability of increasing or decreasing in the child. We first populated the threshold with arbitrary values, and over the course of the game, it would evolve to its environment. We then mapped the values of the thresholds that showed up, but the data was noisy and took a long time to

produce, as many trials resulted in species dying out entirely. This exposes the limitations of this strategy. Because the game does not allow for second chances, the randomness introduced in this sort of evolution is unreliable. Furthermore, since the evolution relies on reproduction, in sparse food environments, the evolution happens too slowly to be successful.

We adapted our strategy to have the evolution happen ahead of time. We decided we would test many different threshold values in many different environments and manually code the successful values into our final organism.

We tested 24 different environments, changing values of p , q , v and u . We left m and n the same, as significantly increasing the size of the environment caused the trials to run for too long. For each environment, we tested all combinations of all three thresholds at values 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, and 100. We also tested with **modesty** turned on and off. We ran each trial for 5000 rounds and counted the number of surviving organisms. For each environment and each threshold configuration, we ran ten trials and took the average number of surviving organisms.

The results provided valuable information for sparse food environments, providing a clear picture of what type of strategies would be successful in survivable.

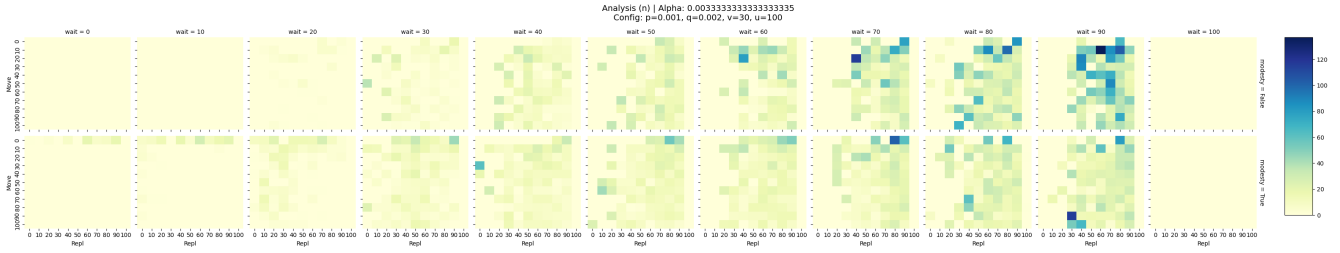


Figure 3: Heatmap of trial results for $p = 0.001$, $q = 0.002$, $v = 30$, $u = 100$. Top row is with **modesty** on and bottom row is with **modesty** off. *replThreshold* mapped on x-axis and *moveTreshhold* mapped on y-axis. *waitTreshhold* increase in graphs from left to right.

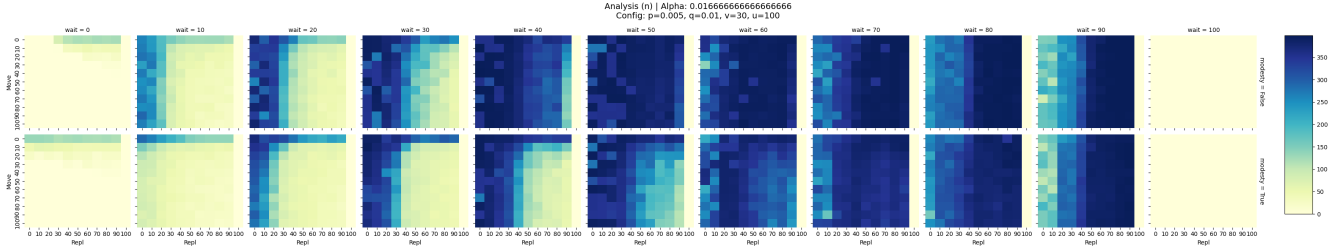


Figure 4: Heatmap of trial results for $p = 0.005$, $q = 0.01$, $v = 30$, $u = 100$.

However, in denser food environments, a large swathe of configurations all resulted in complete population saturation, making the data less valuable. Regardless, it provided a good baseline for what configurations would be successful in solo play.

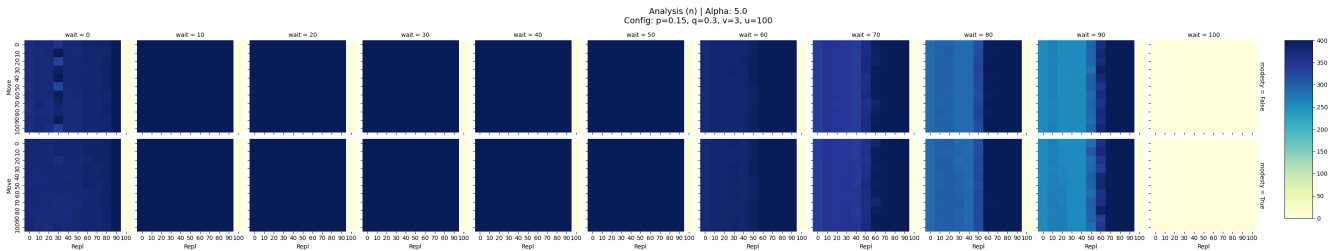


Figure 5: Heatmap of trial results for $p = 0.15$, $q = 0.3$, $v = 3$, $u = 100$.

4.3 Environmental Estimation Module

A central component of our organism is the ability to estimate the effective food spawn probability of the environment from purely local observations. Since the simulator does not provide direct access to the parameter p , the organism must infer it online by observing how nearby empty cells evolve over time. This estimate is later used to determine the current environment category and to adjust internal behavioral thresholds accordingly.

Our implementation is based on a simple trial-based estimator that measures how often food appears in neighboring cells that were previously both empty and food-free. The main design goal is to isolate genuine environmental food generation from changes caused by other organisms. For this reason, we only count observations that remain valid across two consecutive rounds.

Measurement Conditions. The estimation process relies on comparing local neighborhood states between consecutive turns. At a given round t , the organism observes its four orthogonal neighbors and identifies cells that satisfy two conditions: they contain no organism and they contain no food. These cells represent valid candidates for observing spontaneous food appearance.

At the next round, the organism checks the same neighboring positions again. If a previously valid candidate cell is still unoccupied, then the observation is considered valid and contributes to the estimator. If food has appeared in that cell, the event is counted as a successful spawn. If the cell became occupied in the meantime, the observation is discarded, since the change may have been caused by another organism rather than the environment itself.

This logic is implemented through the stored variables `prevMove` and `prevNeighStates`, which allow the organism to compare the current neighborhood with the neighborhood observed in the previous turn. In practice, the estimator is updated only when the previous action preserved the relevance of the local comparison, namely after `STAY_PUT` or `REPRODUCE`. This ensures that the organism is still reasoning about the same local configuration when evaluating the outcome.

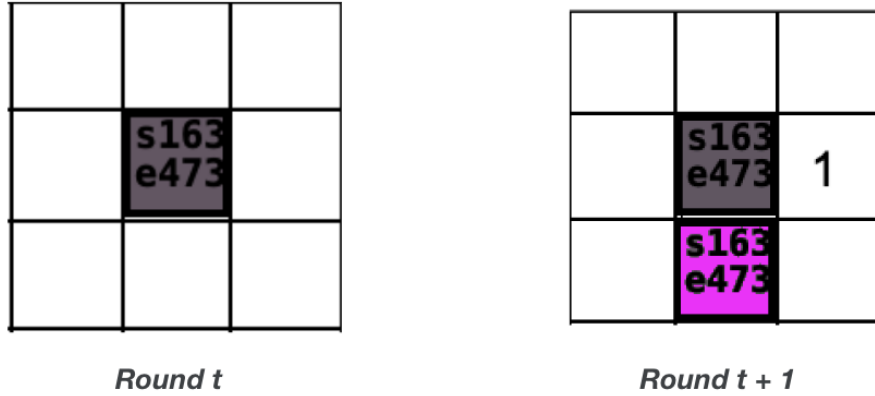


Figure 6: *Example of a single update step in the trial-based food spawn estimation process. Three neighboring cells are empty and food-free at round t and remain unoccupied at round $t + 1$, contributing three valid trials. Food appears in one of them, producing a single hit.*

At round $t + 1$, the same cells are inspected again. Because all three remain unoccupied, they contribute valid trials. Food appears in one of the cells, which is counted as a hit.

This observation updates the running counters by adding three trials and one hit. The probability estimate is then refined using all accumulated observations over time rather than relying on a single measurement.

Trials and Hits. The estimator is maintained through two counters:

- **trials**, the number of valid empty-cell observations;
- **hits**, the number of those observations in which food appeared in the next round.

More precisely, for each neighboring cell observed at round t :

- if the cell was empty and had no food at round t , and remains empty at round $t + 1$, then it contributes one valid trial;
- if, in addition, food is present at round $t + 1$, then it also contributes one hit;
- if the cell becomes occupied at round $t + 1$, it is ignored.

This distinction is important because it prevents the organism from confusing actual food spawning with interference caused by movement or reproduction of nearby organisms.

Probability Estimation. Using the counters above, the organism computes an online estimate

$$\hat{p} = \frac{\text{hits}}{\text{trials}}.$$

This value approximates the probability that food appears in an empty neighboring cell. The estimator is intentionally simple: rather than fitting a more complex model, we use a lightweight statistic that can be updated incrementally at every turn with minimal computational cost.

Although $\hat{p}$ is only a local approximation, repeated observations across time allow the organism to gradually build a useful picture of environmental richness. In sparse environments, the estimate converges more slowly, while in richer environments evidence accumulates more quickly.

Bucket Mapping. Because exact numerical estimates are sensitive to noise, the organism does not use $\hat{p}$ directly for decision-making. Instead, the estimate is mapped into one of five discrete environment buckets:

EXTRA_LOW, VERY_LOW, LOW, MODERATE, HIGH.

The corresponding thresholds are:

$$\begin{aligned} \hat{p} < 0.005 &\rightarrow \text{EXTRA_LOW}, & \hat{p} < 0.01 &\rightarrow \text{VERY_LOW}, & \hat{p} < 0.05 &\rightarrow \text{LOW}, \\ \hat{p} < 0.10 &\rightarrow \text{MODERATE}, & \text{otherwise} &\rightarrow \text{HIGH}. \end{aligned}$$

This discretization improves robustness by preventing small fluctuations in $\hat{p}$ from causing unstable changes in behavior. The resulting bucket is later used by the threshold controller to select appropriate movement, reproduction, and waiting parameters.

Bias in Competitive Settings. The estimator described above assumes that discarded observations are independent of food appearance. However, this assumption no longer holds in competitive environments where other organisms actively move toward food.

Figure 7 illustrates a simplified example of this phenomenon. At round t , the organism observes four empty neighboring cells. Suppose the true environmental probability of food appearance is $p = \frac{1}{4}$, meaning that, on average, one of the four cells will contain food at the next round.

Between rounds t and $t + 1$, a competing organism may move into the cell where food has appeared and consume it before it is observed. As a result, that cell becomes occupied and is discarded from the estimation process. The organism therefore records only three valid trials and zero hits, producing an observed estimate

$$\hat{p} = \frac{0}{3} = 0,$$

which is significantly lower than the true probability.

This phenomenon introduces a systematic *censoring bias*. Cells that become occupied are removed from the sample, yet occupation is not random: competing organisms are more likely to move toward cells containing food. Consequently, discarded observations are more likely to correspond to potential hits rather than misses.

Formally, let T be the set of candidate cells at round t , and let $C \subseteq T$ denote the cells that remain unoccupied at round $t + 1$. The estimator is computed using only cells in C . However, if

$$P(\text{cell occupied} \mid \text{food appears}) > P(\text{cell occupied} \mid \text{no food}),$$

then the filtering process removes a disproportionate number of positive events. As a result,

$$E[\hat{p}] < p,$$

and the estimator becomes biased downward.

Intuitively, the organism tends to lose precisely those observations that are most informative for detecting food generation, while retaining many uninformative observations where no food appeared. This asymmetry causes the estimator to systematically underestimate environmental richness in competitive settings.

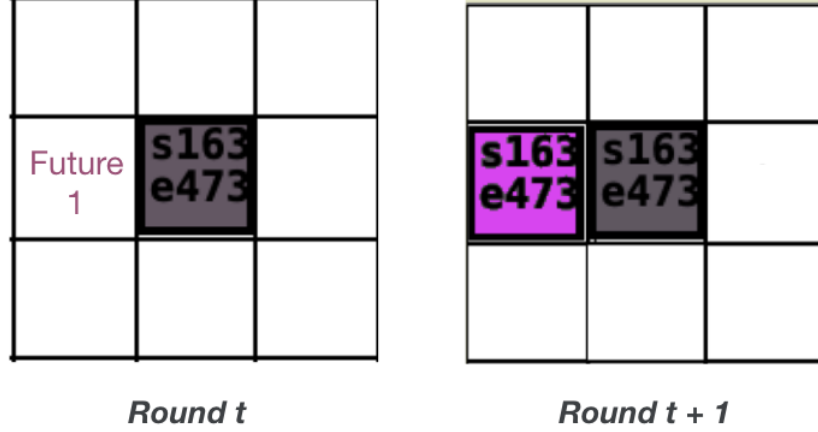


Figure 7: *Censoring bias in competitive environments. A cell where food appears becomes occupied by an opponent before observation and is therefore discarded, reducing the number of trials while also removing a likely hit.*

Confidence Estimation. Because the estimator relies on accumulated observations, its reliability increases with the number of valid trials. Estimates based on limited data are more uncertain, while those supported by many observations are more stable.

In our implementation, confidence is represented as a discrete score in the range $[0, 3]$, computed as a scaled function of the number of trials:

$$\text{confidence} = \min\left(3, \left\lfloor \frac{\text{trials}}{8} \right\rfloor\right).$$

This mapping provides a simple monotonic relationship between evidence and reliability while keeping the representation compact for communication.

The chosen scaling factor is heuristic and was selected to balance responsiveness and stability in practice. Determining an optimal confidence mapping remains an open question that could be explored through more systematic statistical modeling.

4.4 Social Communication and Consensus

While local estimation provides useful information about the environment, it remains limited by partial observability and noise. To improve reliability, organisms exchange compact summaries of their internal estimates through the simulator’s external state mechanism. This enables a distributed consensus process in which multiple organisms collectively refine their understanding of environmental conditions.

External State Encoding. Each organism continuously broadcasts a compact 8-bit message that encodes three types of information:

- a species identifier (**TAG**) used to distinguish allied organisms from opponents,
- the locally estimated environment bucket,
- a confidence value indicating the reliability of the estimate.

The bit layout is structured as follows:

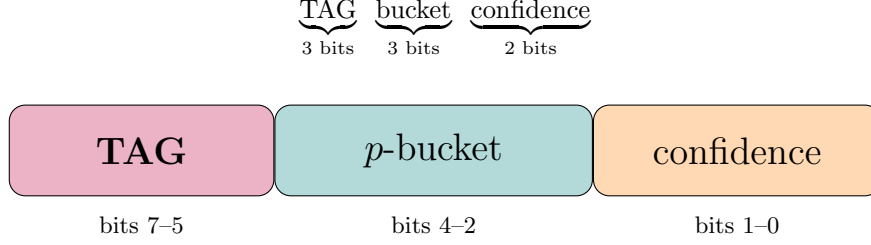


Figure 8: *External state encoding format.* The 8-bit message contains a species identifier (TAG), the estimated environment bucket, and the confidence score.

The species tag occupies the most significant bits and enables organisms to recognize trusted peers. The environment bucket encodes the discretized estimate of food spawn probability, while the confidence field represents the amount of supporting evidence accumulated through local observations.

Message Filtering and Plausibility Checks. Incoming external states are processed through a sequence of validation rules before being incorporated into the consensus mechanism.

1. **Species Tag Verification.** The most significant bits encode a species identifier (TAG). If the received tag does not match the organism’s own identifier, the message is rejected. Such messages are interpreted as originating from opponents and are discarded.
2. **Bucket Validity Check.** The bucket field is encoded using three bits; however, only values in the range $[0, 4]$ correspond to valid environment categories. If the decoded value falls outside this range, the message is considered invalid and discarded.
3. **Plausibility Filtering.** When sufficient local evidence has been accumulated, the organism evaluates the consistency of external estimates with its own. Let b_{local} be the locally computed bucket and b_{ext} the received bucket. If

$$|b_{local} - b_{ext}| > 2,$$

the estimate is considered implausible. In this case, the associated confidence value is reduced by two units to limit its influence on the voting process. If this reduction lowers confidence to zero, the message is discarded entirely.

If any validation step fails, the decoding process returns a null value. This outcome is treated as an indication of an opponent or corrupted information and is used to raise an internal competition flag.

Memory of External Observations. To aggregate information over time, each organism maintains a bounded *External Information Buffer* that stores previously validated external estimates. Rather than relying solely on immediate neighbor interactions, decoded messages that pass the filtering stage are inserted into this buffer as pairs of (bucket, confidence).

The buffer is implemented as a fixed-size queue, ensuring constant memory usage while preserving the most recent observations. When capacity is exceeded, the oldest entries are discarded.

This structure allows the organism to integrate information collected across multiple encounters and time steps, producing a more stable consensus that is less sensitive to short-term fluctuations or sporadic noisy observations.

Weighted Majority Voting. Let

$$B = \{\text{EXTRA_LOW}, \text{VERY_LOW}, \text{LOW}, \text{MODERATE}, \text{HIGH}\}$$

denote the set of the five possible environment buckets.

Each validated external observation contributes a pair

$$(\beta_j, c_j),$$

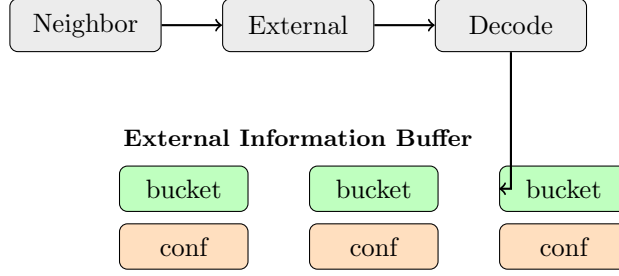


Figure 9: Representation of the external information buffer. Valid decoded messages are stored as pairs of bucket and confidence values.

where $\beta_j \in B$ is the transmitted bucket and $c_j \in \{0, 1, 2, 3\}$ is its confidence score. The organism also contributes its own local estimate $(\beta_{\text{loc}}, c_{\text{loc}})$.

For each bucket $\beta \in B$, we define its total support as

$$S(\beta) = (c_{\text{loc}} + 1) \mathbf{1}\{\beta_{\text{loc}} = \beta\} + \sum_{j=1}^m (c_j + 1) \mathbf{1}\{\beta_j = \beta\},$$

where m is the number of stored external observations and $\mathbf{1}\{\cdot\}$ denotes the indicator function.

The final bucket is then selected according to the maximum-support rule

$$\beta^* = \arg \max_{\beta \in B} S(\beta).$$

Thus, consensus is obtained by aggregating confidence-weighted votes over the five admissible buckets. Observations with higher confidence contribute more strongly, while low-confidence estimates have only limited influence on the final decision.

4.5 Behavioral Threshold Controller

Once an environment bucket has been selected, the organism translates this coarse environmental estimate into concrete behavioral parameters. This mapping is implemented by the routine `setPDependents`, which assigns values to three internal thresholds: `waitThreshold`, `replThreshold`, and `moveThreshold`. These parameters determine whether the organism should conserve energy, reproduce, or actively explore.

The role of this controller is to convert a qualitative estimate of environmental richness into a quantitative behavioral regime. Environments with very low food availability require conservative behavior, since unnecessary movement or aggressive reproduction may quickly deplete energy and lead to extinction. In contrast, richer environments support more aggressive exploration and lower reproduction thresholds, allowing organisms to exploit abundant resources more efficiently.

The threshold controller therefore acts as the main interface between environmental estimation and action selection. Rather than changing the underlying decision logic, the organism keeps the same behavioral structure and only modifies the numerical conditions under which specific actions become favorable. This preserves a unified decision process while still enabling substantial adaptation across environments.

Bucket-Dependent Thresholds. The mapping implemented in our controller is summarized in Table 1. The values were chosen empirically to reflect different survival requirements across scarcity regimes.

Interpretation. The table reveals a clear progression from conservative to aggressive behavior as the estimated environment becomes richer. In the `EXTRA_LOW` bucket, the organism initially remains almost entirely passive, with a very high waiting threshold, a prohibitive reproduction threshold, and zero movement probability. This reflects an extreme survival regime in which preserving energy is more important than expansion.

As the estimated environment improves from `VERY_LOW` to `LOW`, the controller gradually lowers waiting and reproduction requirements, while still keeping exploration relatively limited. In `MODERATE` and `HIGH` environments,

Table 1: Behavioral thresholds selected by the controller as a function of the environment bucket and age. Thresholds are expressed as percentages.

Bucket	Age Range	waitThr	replThr	moveThr
EXTRA_LOW	age < 300	95	110	0
EXTRA_LOW	$300 \leq \text{age} < 600$	70	90	30
EXTRA_LOW	age ≥ 600	70	80	40
VERY_LOW	age < 150	30	30	40
VERY_LOW	age ≥ 150	40	60	15
LOW	age < 150	20	20	40
LOW	age ≥ 150	40	45	15
MODERATE	all ages	10	20	60
HIGH	all ages	0	10	80

the controller shifts decisively toward aggressive behavior: waiting becomes rare, reproduction requires less stored energy, and movement becomes much more likely.

Interaction with Age. For scarce environments, threshold selection also depends on age. This creates a phased control mechanism: younger organisms are initially more exploration-oriented in some low-resource settings, but later transition toward more conservative behavior. In the most hostile regime (**EXTRA_LOW**), the opposite pattern appears: early behavior is highly restrictive, while later phases slightly relax movement and reproduction constraints if the organism has survived long enough.

Competition Override. In addition to bucket-based adaptation, the controller includes a simple competition-aware override. If an opponent is detected through invalid or mismatched external information, the internal variable `enemyDetected` is raised, and the waiting threshold is forced to zero. This suppresses passive waiting under competitive pressure and encourages more active responses in contested settings.

4.6 Decision Engine

The decision engine is responsible for selecting the organism’s action at each simulation step. It integrates environmental information, behavioral thresholds, and current internal state variables into a prioritized rule-based process. Rather than relying on a single scoring function, the organism follows a hierarchy of decision rules that reflect survival priorities.

Priority Structure. Actions are evaluated in a fixed priority order, ensuring that urgent survival behaviors override exploratory or opportunistic actions. The priority hierarchy is as follows:

1. **Food Consumption.** If food is present in an adjacent cell, the organism immediately moves to consume it. This rule has absolute priority, as energy acquisition is essential for survival and future reproduction.
2. **Reproduction.** If no immediate food is available, the organism evaluates reproduction opportunities. Reproduction is permitted only if sufficient energy is available and an adjacent empty cell exists. The required energy level is determined by the reproduction threshold set by the behavioral controller.
3. **Emergency Search.** If energy falls below a critical level, the organism enters an emergency mode that prioritizes locating food. In this mode, the organism actively searches neighboring cells and prefers moves that increase the likelihood of survival, even if they do not align with normal exploration policies.
4. **Exploration.** When no urgent survival action is required, the organism may enter an exploration phase. The probability of exploring depends on the movement threshold determined by the environment bucket. Exploration attempts to move into empty neighboring cells, allowing the organism to discover new food sources and expand spatial coverage.
5. **Passive Waiting.** If none of the above conditions are satisfied, the organism remains stationary. Waiting conserves energy and is favored in resource-scarce environments or when age-dependent thresholds promote conservative behavior.

Energy-Dependent Constraints. Energy level plays a central role in decision-making. Reproduction requires energy above a threshold proportional to the maximum energy capacity, ensuring that replication does not endanger survival. Conversely, very low energy activates emergency behavior, overriding normal exploration policies.

Exploration Mechanism. Exploration decisions are probabilistic and depend on the movement threshold configured by the behavioral controller. When exploration is triggered, the organism samples neighboring cells and selects a feasible empty position. If multiple options are available, the choice is randomized to promote spatial diversity and reduce deterministic congestion.

This stochastic component prevents predictable movement patterns and improves overall grid coverage.

Integration with Threshold Controller. The behavioral thresholds determined by the environment bucket directly modulate the decision engine. The reproduction threshold influences eligibility for replication, the movement threshold controls exploration likelihood, and the waiting threshold biases the organism toward passive behavior.

Thus, the decision engine does not operate independently but executes policies parameterized by the adaptive threshold controller.

Design Rationale. A rule-based priority system was chosen over global optimization methods due to its robustness, interpretability, and low computational cost. The hierarchical structure ensures that essential survival behaviors are never suppressed by secondary objectives, while still allowing flexible adaptation through threshold tuning.

4.7 Lifecycle and Population Management

In addition to environmental adaptation, organism behavior is modulated by lifecycle-dependent controls that regulate population growth and long-term stability.

Age Tracking and Phasing. Each organism maintains an internal age counter that is incremented at every simulation step. Age determines lifecycle phase, which in turn influences behavioral thresholds selected by the controller. Different age intervals correspond to distinct parameter regimes, enabling phased behavioral adaptation without altering the underlying decision logic.

Early vs Late Behavior. Early-stage organisms prioritize expansion and resource acquisition to establish a stable presence. As age increases, behavior gradually shifts toward conservation, reducing unnecessary movement and limiting aggressive reproduction. This transition improves survival probability and prevents late-stage energy depletion.

Offspring Limits and Population Control. Reproduction is constrained by explicit offspring limits to prevent uncontrolled population growth. These limits reduce overcrowding and mitigate resource exhaustion in dense regions. By restricting replication frequency, the organism maintains a sustainable population size across long simulations.

Integration with Behavioral Thresholds. Lifecycle-dependent parameters directly influence the threshold controller, allowing age to modulate movement, waiting, and reproduction tendencies. This integration ensures that both environmental conditions and organism maturity jointly determine behavior.

4.8 Knowledge Inheritance Mechanism

To preserve environmental knowledge across generations, organisms transfer compressed internal statistics through a compact DNA encoding scheme. This mechanism enables offspring to inherit partial estimation state from their parent, avoiding the need to restart learning from an uninformed state.

DNA Bit Encoding. The DNA is represented as a 32-bit integer partitioned into fixed-size bit fields:

$$\underbrace{\text{trials}}_{12 \text{ bits}} \quad \underbrace{\text{hits}}_{12 \text{ bits}} \quad \underbrace{\text{age}}_{8 \text{ bits}}$$

- **Trials (bits 31–20).** Stores the number of valid environmental observations used for probability estimation. The value is capped at $2^{12} - 1 = 4095$ to fit the allocated bit range.
- **Hits (bits 19–8).** Stores the number of successful food-spawn observations. This value is also capped at 4095.
- **Age (bits 7–0).** Encodes the lifecycle stage of the organism, capped at $2^8 - 1 = 255$.

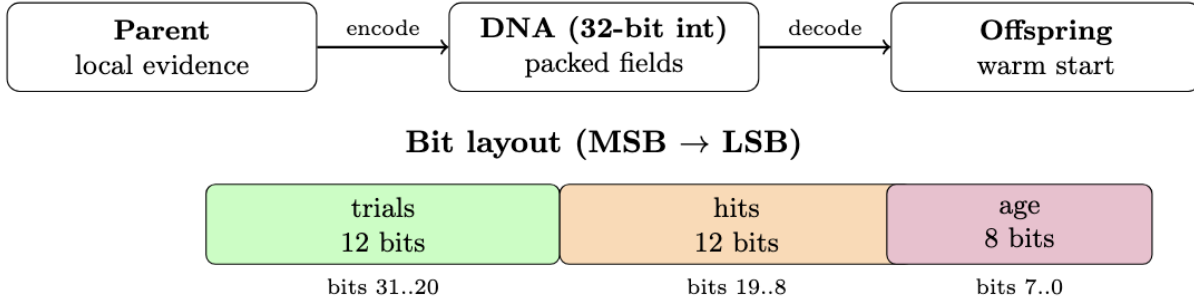
The packed representation is constructed using bitwise shifts:

$$\text{DNA} = (\text{trials} \ll 20) \mid (\text{hits} \ll 8) \mid \text{age}.$$

Transfer and Restoration. During reproduction, the parent encodes its current internal statistics and transmits the resulting DNA to the offspring. Upon initialization, the offspring decodes the bit fields using bit masking and right shifts to restore the inherited state variables:

$$\text{trials} = (\text{DNA} \gg 20) \& 0\text{xFFF}, \quad \text{hits} = (\text{DNA} \gg 8) \& 0\text{xFFF}, \quad \text{age} = \text{DNA} \& 0\text{xFF}.$$

Warm-Start Estimation. By restoring accumulated statistics, offspring organisms resume probability estimation with previously gathered evidence rather than starting from zero. This significantly reduces early-life uncertainty and allows more informed decisions immediately after birth.



Deque / neighbor opinions are **not inherited**
offspring rebuilds them via External State

Figure 10: *DNA-based knowledge inheritance.* The parent encodes local environmental statistics into a 32-bit integer that is decoded by the offspring to obtain a warm start. The bit layout shows the allocation for trials, hits, and age from most significant to least significant bits.

5 Results

5.1 Individual Results and Analysis

Environment	Avg Count (End)	Avg Energy (End)	# Extinctions	Avg Extinction Time
Hot Sands	19.52 (4th)	2842.03 (4th)	72 (5th)	36.21 (8th)
Desert	35.55 (1st)	11885.51 (1st)	54 (3rd)	495.37 (4th)
Monday	182.74 (2nd)	24465.64 (3rd)	0 (7-way tie 1st)	NaN (7-way tie 1st)
Default	174.28 (2nd)	36169.95 (1st)	0	NaN
Rainforest	332.09 (6th)	65447.68 (7th)	0	NaN
Big Food	229.21 (6th)	47575.28 (4th)	0	NaN
Finding Gold	1901.99 (1st)	837113.98 (1st)	2 (3-way tie 3rd)	433.00 (3rd)
Big	4686.84 (1st)	1167603.92 (1st)	0	NaN

Table 2: Performance of FinalBrain across environments (extended trials)

Table 2 summarizes the performance of our FinalBrain implementation across all tournament environments. Each environment modifies key parameters such as food spawn probability (p), food reproduction probability (q), movement cost (v), board size, and energy constraints. These parameters fundamentally change the nature of the survival problem: some environments reward energy conservation, others reward aggressive exploration, and others primarily test scalability as the population grows. Cells without ranking indicate that all groups had the same value. Our strategy was built around three core behaviors: energy-aware movement, replication thresholds to prevent premature reproduction, and modesty constraints that discourage overcrowding, as discussed above. Overall, the results show that these design choices worked well in environments where exploration and energy efficiency were the primary challenges, though they were less effective (compared to other teams) in scenarios that required constant movement.

Sparse Environments: Desert and Finding Gold The *Desert* configuration represents one of the most resource-constrained scenarios. Food appears extremely rarely ($p = 0.001$) and movement is expensive ($v = 30$), meaning that unnecessary exploration rapidly drains energy. In this environment FinalBrain performed particularly well. The implementation achieved the **highest average final population among all groups** and ranked **third** in extinction count, with an average extinction time of roughly 495 ticks. From this data, we hypothesize that due to some level of randomness (i.e. our organisms happen to miss the scarce food that is on the board), we may die out fairly quickly. However, when we do happen to survive, we survive incredibly well.

This outcome aligns closely with the intended design of our strategy. The energy-aware movement rules discourage organisms from moving when energy reserves are low or when no clear opportunity for food exists. In *Desert*, where movement is extremely costly, this behavior prevents organisms from wasting energy searching for food that is unlikely to exist nearby. After detecting a low p value in the beginning, our organism will severely limit reproduction and movement until around timestep 300, where adequate food likely has spawned on the board. Organisms conserve energy and survive long enough to exploit the rare food spawn events, and as a result, the conservative movement logic performed exactly as intended.

The *Finding Gold* environment presents a different challenge. Food appears even more rarely ($p = 0.0005$), but movement is inexpensive ($v = 5$) and the board size increases to 50×50 . In this setting survival depends on both conserving energy and on effectively exploring the grid to "find gold." FinalBrain performed well here, reaching an average population of approximately 1900 organisms with only two extinctions.

Although we had not designed a strategy specifically for this environment, and did not explicitly consider v in our decision logic, our *Desert* strategy was likely employed and proved highly effective. The *Desert* strategy was designed for extremely low food probabilities and therefore emphasizes cautious reproduction, conservative energy usage, and exploration when food cannot be found locally. These behaviors translate well to the *Finding Gold* configuration: low reproduction thresholds prevent rapid population collapse before food is discovered, while increased exploration in later stages allows organisms to spread across the larger grid and locate rare food sources once they have accumulated. Additionally, our population control mechanisms (such as limiting reproduction per organism and maintaining relatively high reproduction thresholds) helped prevent overpopulation even after food

was found. As a result, the strategy effectively balanced energy conservation with exploration, which is the key requirement for success in this environment.

Environments that Penalize Stationary Behavior: Hot Sands The *Hot Sands* environment introduces a fundamentally different constraint: remaining stationary incurs a large energy penalty ($s = 20$), whereas moving is less costly ($v = 10$, as in the default). Organisms must continue moving in order to survive. This environment proved to be the most difficult for FinalBrain, which ranked **fifth** in extinction count and **eighth** in extinction time.

This result exposes a limitation in our approach. Because the strategy was designed to conserve energy through cautious movement, organisms occasionally remained stationary since doing so in most cases would be energy-efficient. In *Hot Sands*, however, staying still is actively harmful, so the energy-conservation heuristic sometimes delayed exploration (and caused high energy losses) when continuous movement would’ve been beneficial for both energy conservation and food detection.

Balanced Environments: Default and Monday The *Default* configuration ($p = 0.01$, $q = 0.02$) represents a balanced environment where both movement and food availability are moderate. In this setting FinalBrain ranked **second** in final population and **first** in final energy with no extinctions.

These results suggest that the replication thresholds and modesty constraints worked effectively. By preventing organisms from reproducing too early or in already crowded regions, the population was able to grow steadily without exhausting local resources. The high final energy values indicate that organisms consistently located food while maintaining sustainable population density.

The *Monday* configuration introduces a lower starting energy ($\text{INIT_ENERGY} = 250$) and a maximum energy cap of $M = 300$. With a lower movement cost ($v = 7$), organisms begin the simulation with a substantial buffer against starvation. In this environment, many teams successfully avoided extinction, resulting in a seven-way tie in survival. However, FinalBrain achieved the second-highest final population while maintaining strong energy levels.

Because survival was largely guaranteed in this environment (resources were sufficient, and competition was lacking), we believe that differences between strategies primarily reflect how efficiently populations expand. The results indicate that our replication thresholds allowed the population to grow steadily without destabilizing the system.

Resource-Rich Environments: Rainforest and Big Food The *Rainforest* environment provides abundant food ($p = 0.1$, $q = 0.2$), while *Big Food* increases the nutritional value of each food unit ($u = 200$). In both environments FinalBrain maintained stable populations with no extinctions.

However, the final population rankings were somewhat lower (sixth place in both environments). This outcome reflects a trade-off in our approach. Because the replication rules intentionally restrict reproduction until sufficient energy reserves exist, the strategy prioritizes stability over aggressive population expansion. In extremely resource-rich environments, more aggressive strategies can sometimes produce larger populations more quickly, which is why we believe our ranking may have been lower in these two scenarios. Nevertheless, our implementation maintained strong energy reserves and avoided collapse, indicating that the modesty and replication thresholds successfully prevented overcrowding and energy depletion, so we still managed to fulfill our goals in these cases.

Large-Scale Environment: Big The strongest performance occurred in the *Big* configuration, where the grid size increases to 100×100 . In this environment exploration and spatial distribution become the dominant challenges. FinalBrain achieved **first place in both final population and final energy**, reaching an average population of nearly 4700 organisms with no extinctions.

This result highlights the scalability of the exploration behavior. Because organisms initially spread out before replicating aggressively, the population was able to occupy large portions of the grid without immediately competing for the same resources. Once the population was distributed across the environment, replication produced rapid expansion without creating severe local resource shortages.

Overall Interpretation Across all environments, FinalBrain had overall high survival rates, but performed best compared to other teams when survival depended on efficient exploration and careful energy management. The

energy-aware movement logic proved particularly effective in environments with high movement costs, while the exploration component allowed the strategy to scale effectively in large environments.

The main weakness of the strategy appeared in environments that penalize stationary behavior, such as Hot Sands, where the conservative movement policy becomes counterproductive. Additionally, the cautious replication rules slightly limited peak population sizes in resource-rich environments.

Despite these limitations, the results demonstrate that the combination of energy-aware movement, exploration, and controlled replication produces a robust strategy capable of adapting to a wide range of environmental conditions.

5.2 Competitive Results

Environment	Avg Count (End)	# Extinctions	Avg Extinction Time
Desert	0.93	195	695.55
Hot Sands	0.00	200	48.16
Finding Gold	0.04	193	494.21
Big Food	0.01	199	1029.19
Rainforest	0.57	134	2904.75
Default	0.19	185	1392.31
Monday	0.07	194	771.07
Big	60.11	45	1901.09

Table 3: Performance of FinalBrain across environments

Table 3 presents the competitive results for the *FinalBrain* strategy relative to other groups across each tournament environment. The table reports three primary metrics: the average final population, the number of extinction events across trials, and the average time when extinction occurred. These metrics allow us to evaluate both the survival reliability of the strategy and its ability to maintain population size under different environmental conditions. The results show that FinalBrain performs strongest in larger environments such as *Big*, while survival becomes more challenging in environments with severe resource scarcity or high movement costs.

Average Survival Duration Across Environments

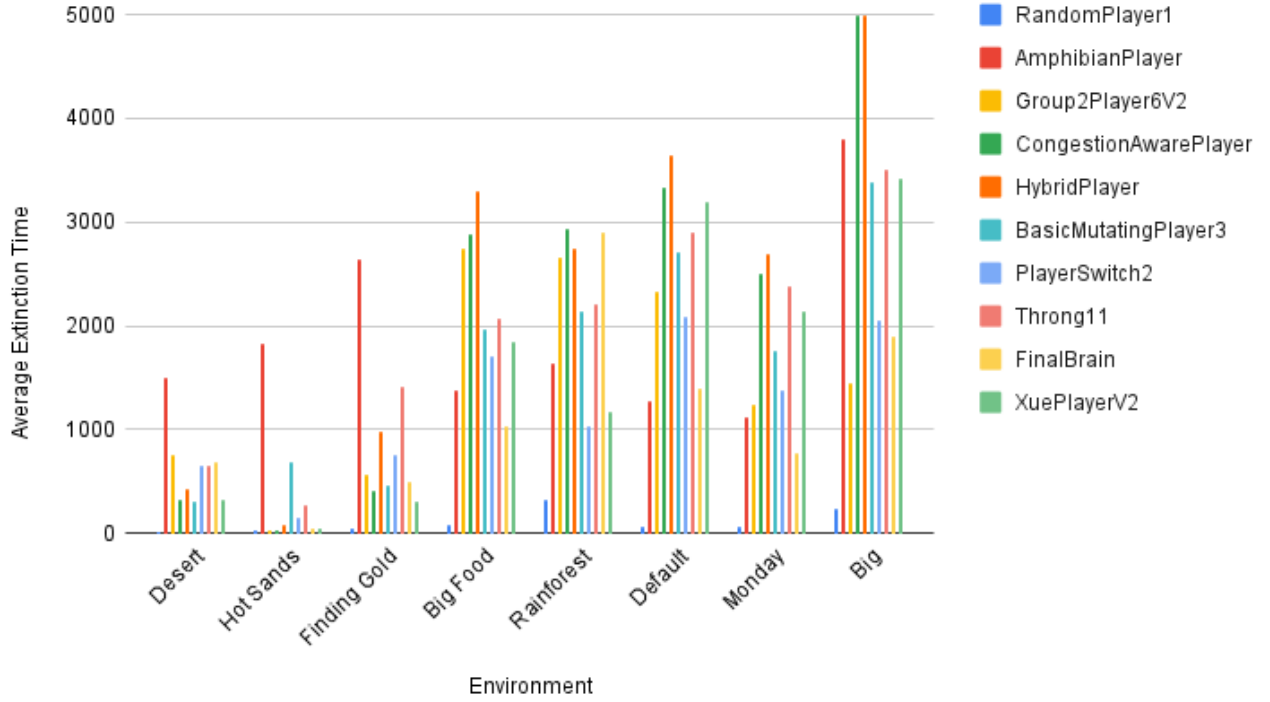


Figure 11: Average extinction time over 200 trials for each environment. Higher values indicate organisms survived longer before extinction.

Figure 8 compares the average extinction time for each competing strategy across all environments. Extinction time measures how long a population survives before collapsing, making it a useful indicator of robustness in difficult environments. Higher values indicate that a strategy was able to sustain its population longer under the given environmental conditions. For players that did not go extinct, values in the graph are set to 5000, representing the length of the game and displaying their success. The figure highlights significant differences between strategies in resource-scarce environments such as Desert and Hot Sands, while many strategies achieve longer survival times in environments with more abundant resources or larger grid sizes.

Group	Total # Extinctions	# Configs with 100% Extinction
1	1157	3
2	866	2
3	1049	2
4	938	1
5	1355	1
6	1110	2
7	1296	1
8	1345	1
9	1355	5

Table 4: Extinction summary across all configurations

Table 4 summarizes extinction outcomes across all tournament environments for each competing group. The first column reports the total number of extinction events observed across all configurations, while the second column records how many environments resulted in complete extinction across all trials. These metrics provide a high-level view of the overall robustness of each strategy. Lower values in both columns indicate strategies that are more

consistently able to sustain populations across different environmental conditions.

Overall Analysis and Interpretation While the individual results in Table 2 show that FinalBrain performs strongly in several environments, particularly in large environments such as *Big*, the competitive results presented in Table 3, Figure 8, and Table 4 reveal a different overall picture when comparing strategies across the competitive tournament.

Prior to the tournament we expected the strategy to perform well in several configurations, particularly environments such as *Desert* and *Default*. In earlier testing and internal trials, our organisms were generally able to survive in these environments for long periods of time, suggesting that the conservative movement and energy management rules were working as intended. However, the competitive results indicate that other groups’ updated strategies were able to outperform our strategy in these same environments.

As shown in Table 4, FinalBrain (Group 8) experienced 1345 extinction events across all configurations, placing it among the higher extinction totals in the tournament. In addition, one configuration resulted in complete extinction across all trials (however, this configuration also resulted in complete extinction for 7/8 of the other teams). Compared to several other strategies with lower extinction totals, this suggests that our implementation was less robust across the full range of environments used in the tournament when considering competition.

Figure 8 helps illustrate where these differences occur. In several environments where resources are scarce or movement constraints are strong, some competing strategies maintain longer survival times before extinction. While FinalBrain often survives for a moderate amount of time, its extinction times are generally lower than those of the most robust strategies in the most difficult environments.

One reason for this outcome lies in the adaptive threshold strategy used by FinalBrain. Our implementation dynamically adjusted behavior based on two key factors: an estimated food probability (referred to as the *P bucket*) and the age of the organism. In extremely low-*p* environments, such as *Desert*, the strategy begins conservatively. Organisms move very little and reproduce rarely in order to conserve energy when food is extremely unlikely to appear. As time passes and organisms begin to die out, the population density decreases and the strategy gradually increases movement in order to search more aggressively for food.

In environments with slightly higher food probabilities, the strategy follows the opposite pattern. Organisms move more aggressively early in their lifespan to capture territory and spread across the grid while resources are still available. As organisms age, movement and reproduction thresholds become more conservative in order to maintain long-term stability.

In theory, this adaptive approach should allow the organism to balance exploration and conservation depending on the environmental conditions. In practice, however, several factors may have limited its effectiveness in competitive play. First, the *P*-bucket estimates are only approximations of the true food generation rates and can be distorted by the presence of competitors consuming nearby food. Also, if the environment behaves differently than expected during early simulation steps, the organism may adopt a suboptimal behavioral mode. Second, the age-based adjustments introduce additional delays before behavior shifts occur, meaning the strategy may react more slowly than simpler policies that aggressively explore from the beginning.

Another factor is population control. FinalBrain intentionally restricts reproduction through energy thresholds and modesty rules in order to avoid overcrowding and resource depletion. While this improves stability in many environments, it can also slow population recovery after losses. Competing strategies that reproduce more aggressively may recover more quickly after temporary population declines, giving them an advantage in competitive environments where rapid expansion is beneficial. As a result, FinalBrain may struggle to gain control over large portions of the board in competitive environments. Furthermore, if board control is lost, the relatively static reproduction thresholds can make it difficult for the population to recover quickly and reestablish its presence.

Overall, the competitive results significantly contrast the conclusions drawn from the individual experiments, but are not contradictory. FinalBrain successfully balances exploration, energy conservation, and population control in many environments, but its adaptive threshold system reacts more slowly than competing strategies. Additionally, our implementation relied on a single core strategy governed by a small set of threshold parameters, while some of the more successful teams appeared to employ multiple specialized strategies that activated depending on the detected environmental conditions. Future improvements would likely focus on making behavioral adjustments more responsive to environmental conditions and allowing replication strategies to adapt more dynamically as the population evolves.

6 Contributions

One of our main contributions to the class discussion was our approach to estimating and communicating environmental parameters, particularly the probability of food spawning (p). Our group implemented a *P-bucketing* approach, where organisms classified the environment into one of several probability buckets ranging from very low to high food availability. While other groups explored similar ideas around environment estimation, we contributed to the discussion around practical ways of computing and transmitting these estimates efficiently.

More broadly, our group focused on a threshold-based strategy that adapted organism behavior based on estimated environmental conditions and organism state (such as age and energy level). Through extensive experimentation and tuning, we explored how movement and reproduction thresholds could be adjusted dynamically to balance exploration and energy conservation across different environments.

Mihnea David. David contributed to the following efforts:

- Developed the initial **BasicPlayerMD1** and **BasicPlayerMD2** models, introducing a structured hierarchical rule-based decision framework that became the foundation of the final organism architecture.
- Proposed and implemented the local environmental estimation mechanism, including the approximation of the food spawn probability p_{local} from neighborhood observations.
- Introduced the discretization of environmental estimates into probability buckets, enabling robust threshold-based behavioral adaptation.
- Proposed the concept of global environmental approximation through distributed information sharing, leading to the design of the external state communication protocol and the consensus mechanism.
- Designed and implemented the DNA-based knowledge inheritance mechanism, enabling intergenerational transfer of environmental statistics and providing offspring with a warm-start estimation state.
- Played an important role in the development of the strategic direction and conceptual design of the project, leading the core implementation process and contributing substantially to both the writing of the final report and the preparation of the presentation.

Vedha Avali. Vedha contributed to the following efforts:

- Conducted manual simulation trials to evaluate the organism's behavior across different environments, identifying weaknesses in the initial parameter settings and guiding subsequent strategy improvements.
- Introduced phase-based behavioral logic that modified organism aggressiveness over time, prioritizing early exploration and reproduction in high-food environments while adopting more conservative behavior in low-food environments.
- Implemented additional reproduction control logic to prevent excessive population growth, helping stabilize energy consumption and reduce the likelihood of rapid extinction due to overpopulation.
- Ran and analyzed competitive simulation trials to evaluate how the organism behaved in multi-species environments, using these results to refine movement and reproduction rules.
- Focused on the *fine-tuning for competitive play* and *results* sections of the presentation, as well as the *Results* and *Future Directions and Limitations* portion of the report.

Evelyn Yu. Evelyn contributed to the following efforts:

- Performed initial analysis and observations of what goals we might try to achieve and calculations to estimate what an ideal organism would look like.
- Established choice of initial thresholds and designed an evolution-based organism to determine successful values.
- Created individual organisms each successful in specific environments in both solo and competitive play.

- Adapted the simulation framework to run faster, automated trials with parametrized organism constructors (instead of using reflection with default constructors) for solo play. Made acquiring data about different thresholds easier.
- Executed parameterized tests to establish baseline for successful thresholds in different solo player environments.
- Tuned final model for competitive play in sparse "desert" environments. Added additional reproduction barriers for the organism in these harsh environments.

7 Future Directions and Limitations

While the FinalBrain strategy demonstrated strong performance in several individual environments, the competitive results revealed a number of limitations in the current approach.

One limitation of our implementation is that it relied primarily on a single adaptive strategy driven by a small set of behavioral thresholds. Although these thresholds were adjusted dynamically based on estimated food probability and organism age, the overall strategy remained largely uniform across environments. In contrast, some higher-performing teams appeared to employ multiple specialized strategies that activated conditionally based on detected environmental characteristics. A case-based approach that selects between fundamentally different behavioral policies may therefore provide better adaptability across diverse environments.

Another limitation is that our strategy did not explicitly incorporate several environmental parameters that were known in advance, such as movement cost (v) or food value (u). For example, when movement costs are low it may be advantageous to adopt a more aggressive exploration strategy, whereas when movement is expensive organisms should conserve energy and avoid unnecessary travel. Similarly, environments where food provides significantly higher energy rewards may benefit from a specialized strategy that initially prioritizes waiting, followed by a controlled exploration phase to efficiently search the grid. Incorporating these parameters directly into the decision-making logic could allow the organism to adopt more appropriate strategies for different environments.

Additionally, the threshold-based system may respond too slowly to changing environmental conditions. Because many behavioral adjustments depend on organism age or gradual threshold changes, the strategy can take time to transition from exploration-focused behavior to energy-conservation behavior or vice versa. More responsive mechanisms, such as monitoring recent food discovery rates or local population density, could allow organisms to adapt their behavior more quickly.

Population control was another area where improvements could be made. While our modesty and replication thresholds helped prevent overcrowding and resource collapse in several environments, they also slowed population recovery after losses. More dynamic reproduction policies that respond to current population size or local resource availability might improve resilience in competitive scenarios.

Overall, the results suggest that combining the existing threshold-based system with additional environment-aware strategies and more responsive adaptation mechanisms could significantly improve the robustness and competitiveness of the FinalBrain approach.

Aside: Runtime Limitations and Scalability During the official tournament, the *Big* and *Finding Gold* environments exhibited significantly longer runtimes and in some cases failed to complete due to out-of-memory exceptions. These environments allow extremely large populations to form. For example, the *Big* configuration uses a 100×100 grid, and *Finding Gold* combines a large board with high food rewards and low movement costs, enabling thousands of organisms to coexist simultaneously. In our individual results, we had an average end count of 1901.99 for *Finding Gold* and 4686.84 for *Big*.

When population sizes grow into the thousands, the organism decision logic, particularly the `searchForNeigh()` and `decidePBucket()` routines, is executed millions of times over a single simulation. In addition, some parts of the neighbor evaluation logic allocate temporary objects during each decision step, which likely increased memory pressure and garbage collection activity, occasionally leading to out-of-memory failures.

We believe that for future scalability, several improvements could mitigate these limitations. First, environmental bucket calculations in `decidePBucket()` could be cached or updated only periodically rather than recomputed every turn. Second, temporary neighbor state allocations could be replaced with reusable structures or primitive arrays

to reduce memory overhead. Finally, simplified decision logic could be used when populations become very large, reducing per-organism computation and improving scalability in large simulations.

8 Acknowledgments

We would like to acknowledge Group 9 for the idea of performing more aggressive exploration when an organism's energy becomes critically low.

We would also like to acknowledge Group 1 for the idea of waiting at the beginning of the game to farm food.

In addition, several ideas discussed during class influenced aspects of our design, particularly the general tradeoff between exploration and conservation in low-resource environments. These discussions helped inform our use of energy thresholds and environment-dependent movement probabilities.

ChatGPT was used to assist with brainstorming and creating an initial draft of certain sections of this report, specifically the Results and Future Directions/Limitations sections. All content was subsequently reviewed, edited, and refined by our team.

9 Conclusion

Overall, this project was interesting and engaging. One aspect we particularly liked was the openness of the assignment. Unlike the previous football tournament project, where everyone was working toward a fixed objective, this project allowed each group to define their own goals and design strategies accordingly.

We also felt that this project served as a good progression from the previous one. It required many of the same skills, such as reasoning about agent behavior and designing algorithms, but in a more complex environment where emergent behavior and unknown environmental parameters played a larger role.

One suggestion for improvement on the project is potentially creating a more robust framework that simulates plants as well as animals. Currently all organisms must simulate behavior most similar to that of animals and for this reason is quite competitive: everyone is looking for the same resources. It could be interesting to create an environment that inherently lends itself to more collaborative play by having the possibility for species (animals) that need other species (plants) to live, and vice versa. Otherwise, the project was very exciting and informative!